

CO2-AT-1

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Course code: CSA1406

Course: Compiler design

1) LL(1) Predictive Parser Design

Given Grammar:

$$S \rightarrow A B \mid D$$

$$A \rightarrow a \mid D b \mid E$$

$$B \rightarrow g \mid D \mid A \mid E$$

$$D \rightarrow e \mid f$$

First

$$\text{FIRST}(S) = \{a, d, e, f, g\}$$

$$\text{FIRST}(A) = \{a, e, f, \epsilon\}$$

$$\text{FIRST}(B) = \{g, d, e\}$$

$$\text{FIRST}(D) = \{e, f\}$$

FOLLOW

$$\text{FOLLOW}(S) = \{\$ \}$$

$$\text{FOLLOW}(A) = \{g, d, e, f, \$ \}$$

$$\text{FOLLOW}(B) = \{e, f\}$$

$$\text{FOLLOW}(D) = \{b, \$ \}$$

ii) Predictive Parsing Table

NT	a	d	e	f	g	\$
S	ABD	ABD	ABD	ABD	ABD	A
A	a	ϵ	D ϵ /E	D ϵ /E	ϵ	ϵ
B	-	DA	ϵ	ϵ	gD	
D	-	-	e	f	-	-

iii) LL (1) verification

FIRST / FIRST conflict: NO

FIRST / FOLLOW conflict: yes ($A \rightarrow D\epsilon$ and $A \rightarrow \epsilon$)

Multiple entries: yes

Conclusion: Grammar is not LL(1)

iv) Predictive Parsing

Stack	Input	Action
S\$	aed\$	$S \rightarrow ABD$
ABD\$	aed\$	$A \rightarrow a$
BD\$	ed\$	$B \rightarrow \epsilon$
D\$	ed\$	$D \rightarrow e$
\$	d\$	Reject

v) Recommendations.

* Grammar cannot be directly used for LL(1)

* cause: FIRST/FOLLOW conflict in A

2) shift-Reduce Parser

Stack	Input	Action
\$	id*id/(id+id)\$	shift id
\$E	*id/(id+id)\$	Reduce
\$E*E	/(id+id)\$	Reduce $E \rightarrow E*E$
\$E/E	(id+id)\$	shift (
\$E/(E)	\$	Reduce (E)
\$E	\$	Accept

Handles.

id

id

$E*E$

id

id

$E+E$

(E)

conflict

* Shift Reduce conflict exists

* Resolved using operator precedence ($*$, $/$ higher than $+$, $-$) and left associativity.

Result : Input Accepted.

3) Recursive Descent Parser

Procedures.

$S()$

$L()$

$L'()$

Functions calls

$S()$

→ match (

→ $L()$

→ $S()$

→ match (a)

→ $L'()$

→ match ()

Result : Input (a) is Accepted.

Error Recovery

- * Report error
- * skip invalid token
- * continue parsing

4) Operator Precedence Parser

Grammar

$$E \rightarrow E+T \mid E-T \mid T$$

$$T \rightarrow T * F \mid F$$

$$F \rightarrow (E) \mid id$$

Precedence

$$() > * > + -$$

Parsing

Stack	Input	Action
\$	id-id*id\$	shift id
\$E	-id*id\$	Reduce
\$E-E*E	\$	Reduce $E * E$
\$E	\$	Reduce $E - E$
\$E	\$	Accept

5) LALR (1) Parser

Augmented Grammar

$$S' \rightarrow S$$

$$S \rightarrow AA$$

$$A \rightarrow aA \mid b$$

FIRST & FOLLOW

$$\text{FIRST}(S) = \{a, b\}$$

$$\text{FIRST}(A) = \{a, b\}$$

$$\text{FOLLOW}(S) = \{\$ \}$$

$$\text{FOLLOW}(A) = \{a, b, \$ \}$$

LALR Table

- * Shift on a, b
- * Reduce using $A \rightarrow b, A \rightarrow aA$
- * Accept on $\$$

Parsing (Input: aabb)

Stack	Input	Action
0	aabb\$	shift a
0a	abb\$	shift a
0aa	bb\$	shift b
0A	b\$	Reduce $A \rightarrow b$
0AA	\$	Reduce $S \rightarrow AA$
0S	\$	Accept

Result:

Input aabb is Accepted.